

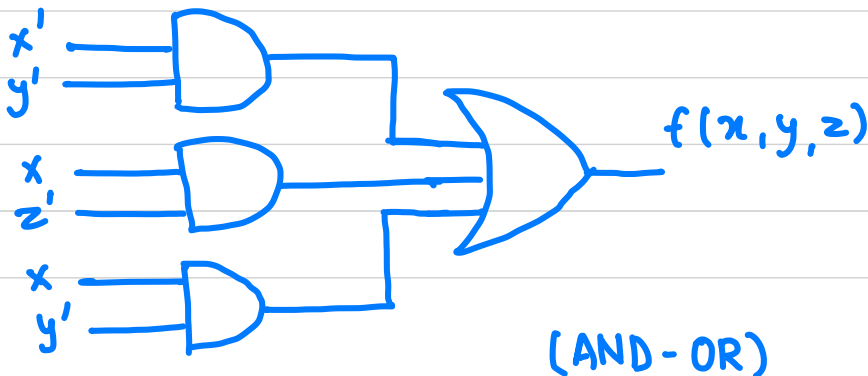
PS-1

1. Decimal value	Binary value	Representation
0	000	$x'y'z'$
1	001	$x'y'z$
4	100	$xy'z'$
5	101	$xy'z$
6	110	xyz'

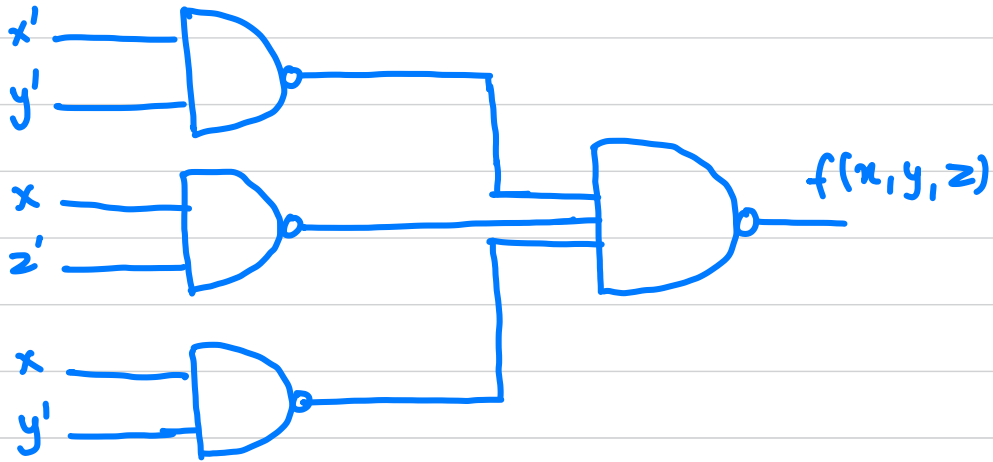
$$f(x, y, z) = x'y'z' + x'y'z + xy'z' + xy'z + xyz'$$

z \ xy	00	01	11	10
0	1	0	1	1
1	1	0	0	1

Minimized function $f(x, y, z) = x'y' + xz' + xy'$



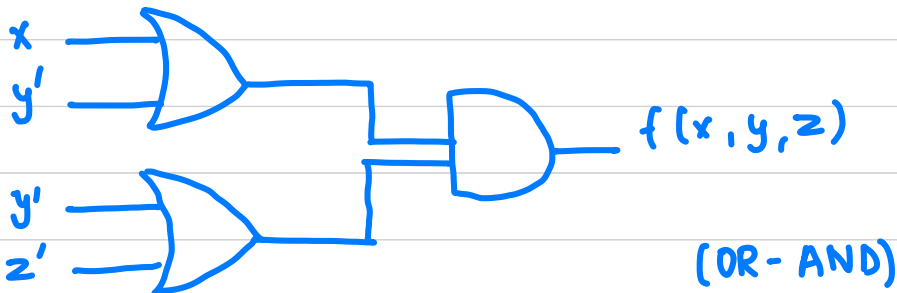
NAND- NAND realization



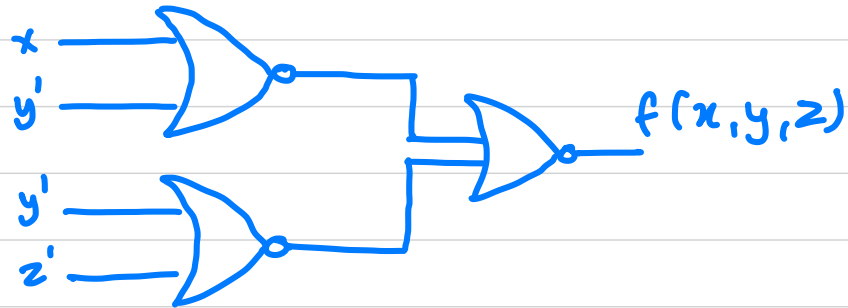
2. $z \backslash xy$

	00	01	11	10
0	1	0	1	1
1	1	0	0	1

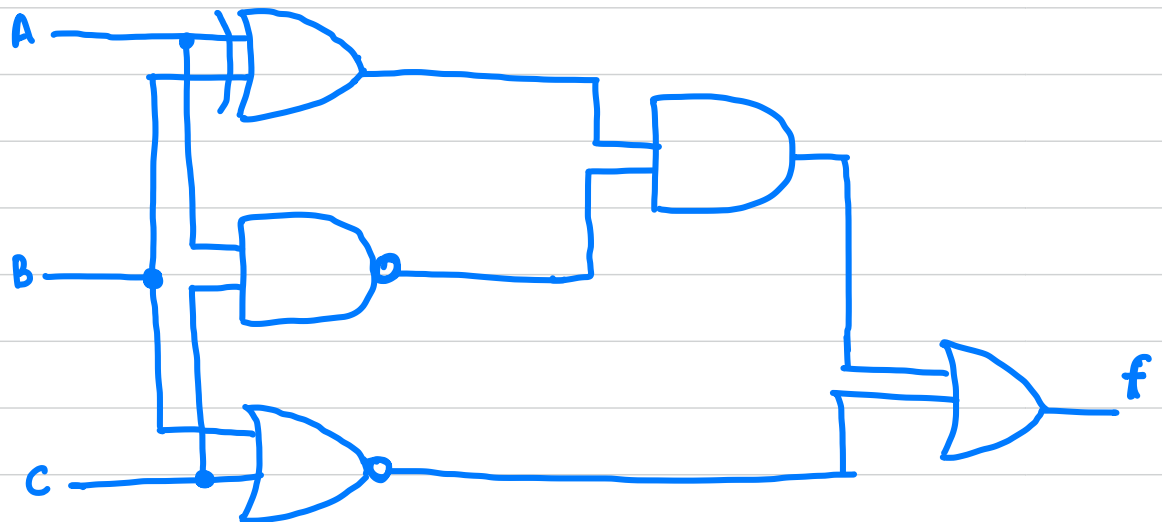
$$f(x, y, z) = (x + y')(y' + z')$$



NOR - NOR realization



3. Given circuit



$$f = (A'B + B'C')$$

		AB			
C		00	01	11	10
	0	1	1		1
	1		1		

$$\text{MSOP} = B'C' + A'B$$

		AB			
C		00	01	11	10
	0			0	
	1	0		0	0

$$\text{MPOS} = (B + C')(A' + B')$$

Worst case delay of the original circuit: $3t_{\text{gate}}$

Worst case delay of the MSOP circuit: $2t_{\text{gate}}$

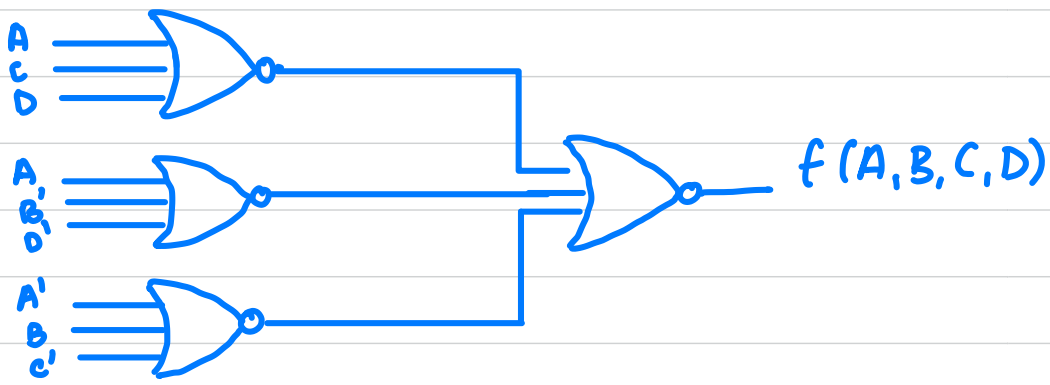
4.

$$f(A, B, C, D) = \sum m(1, 2, 3, 5, 6, 7, 8, 9, 12, 14)$$

$$= \prod M(0, 4, 10, 11, 13, 15)$$

		AB			
CD	AB	00	01	11	10
	00	0	0	1	1
01	01	1	1	0	1
11	11	1	1	0	0
10	10	1	1	1	0

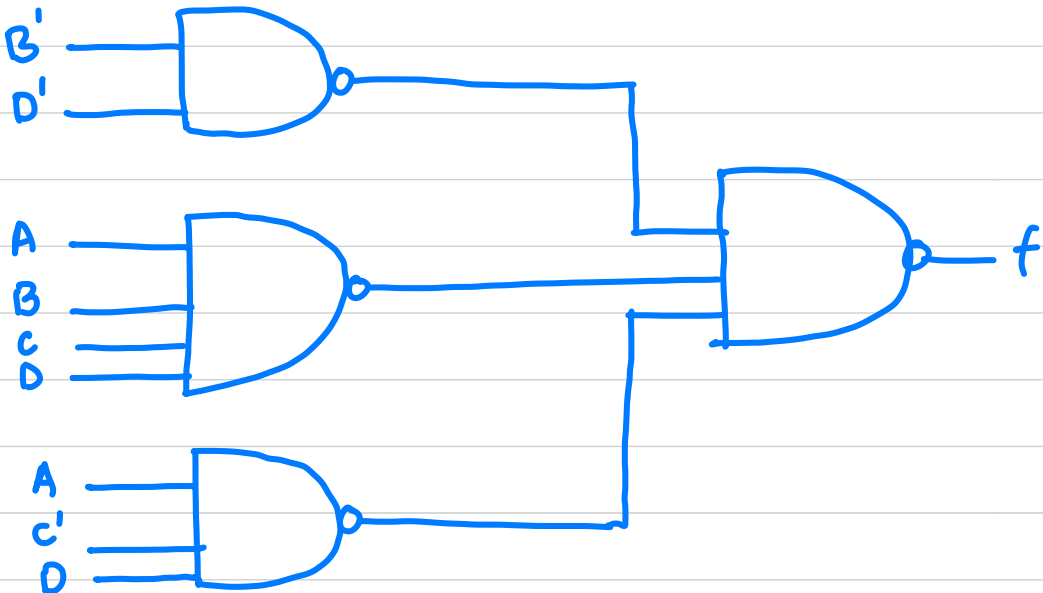
$$f(A, B, C, D) = (A + C + D)(A' + B' + D')(A' + B + C')$$

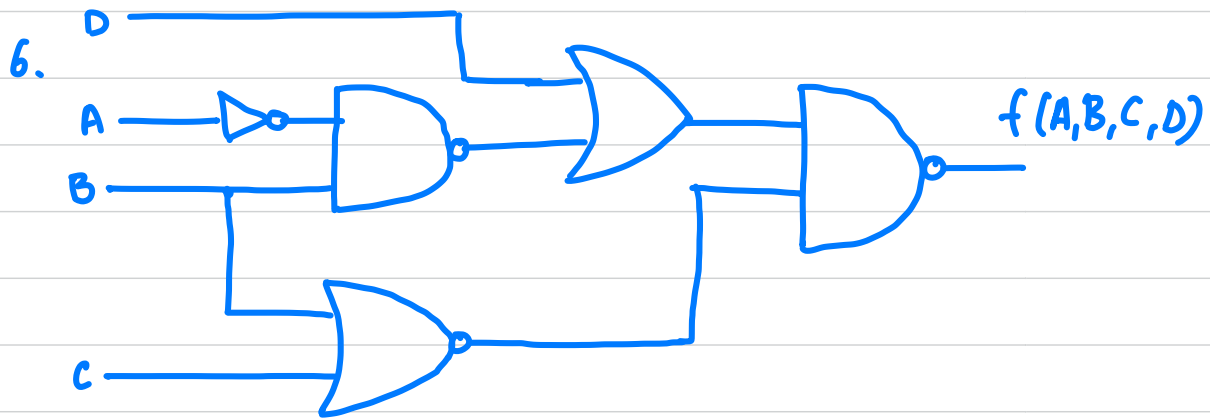


5. $f(A, B, C, D) = \sum m(0, 2, 8, 10, 12, 15)$

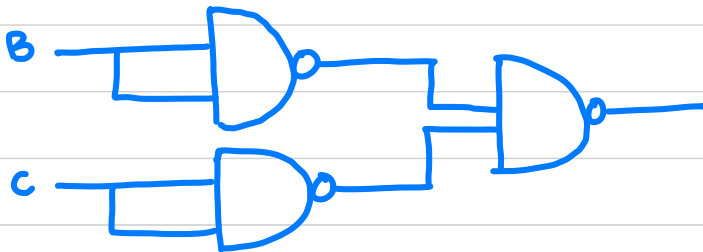
CD \ AB	AB			
	00	01	11	10
00	1	0	1	1
01	0	0	0	0
11	0	0	1	0
10	1	0	0	1

$$f = B'D' + ABCD + AC'D'$$

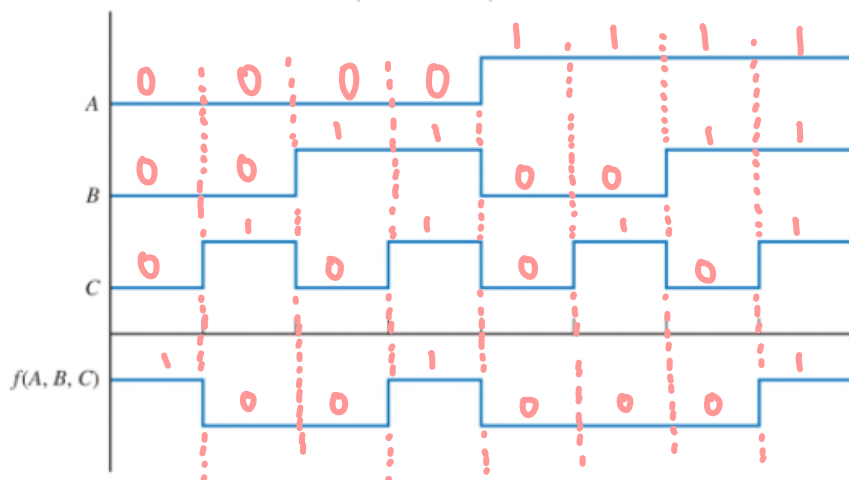




$$\begin{aligned}
 F &= \overline{(\overline{A'B} + D)(\overline{B+C})} \\
 &= \overline{\overline{A'B} + D} + \overline{\overline{B+C}} \\
 &= (\overline{\overline{A'B}}) \cdot \overline{D} + B + C \\
 &= \overline{A}BD + B + C \\
 &= B(\overline{A}D + 1) + C \\
 &= B + C
 \end{aligned}$$



7.

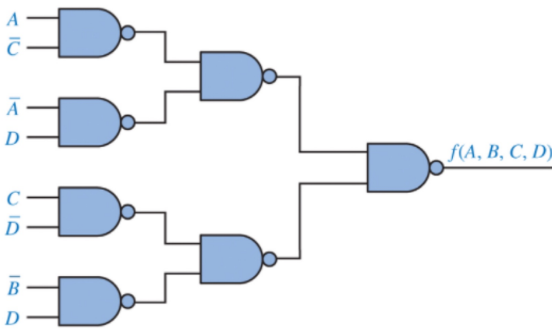


$$F(A, B, C) = A'B'C' + A'BC + ABC$$

		AB			
C		00	01	11	10
	0	1			
	1		1	1	

$$f(A, B, C) = A'B'C' + BC$$

8.



$$F = (\overline{A\bar{C}})(\overline{\bar{A}D}) \cdot (\overline{C\bar{D}}) \cdot (\overline{\bar{B}D})$$

$$= (\overline{A\bar{C}})(\overline{\bar{A}D}) + (\overline{C\bar{D}}) \cdot (\overline{\bar{B}D})$$

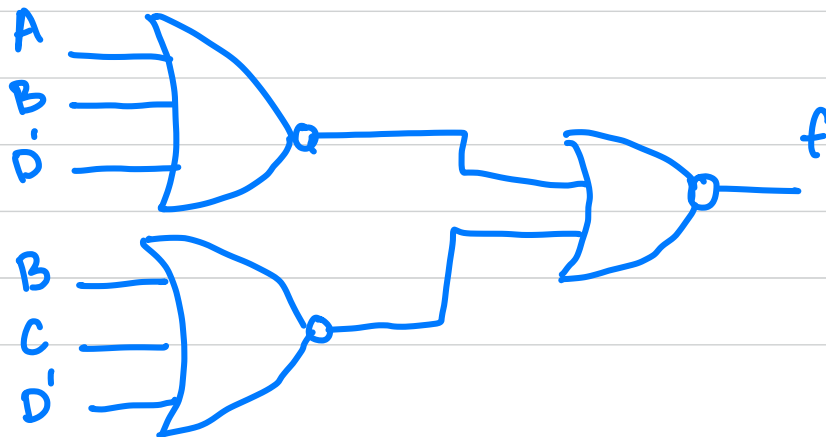
$$= (\bar{A} + C)(A + \bar{D}) + (\bar{C} + D)(B + \bar{D})$$

$$= (\cancel{\bar{A}A} + \bar{A}\bar{D} + CA + C\bar{D}) + (\bar{C}B + \bar{C}\bar{D} + DB + \cancel{D\bar{D}})$$

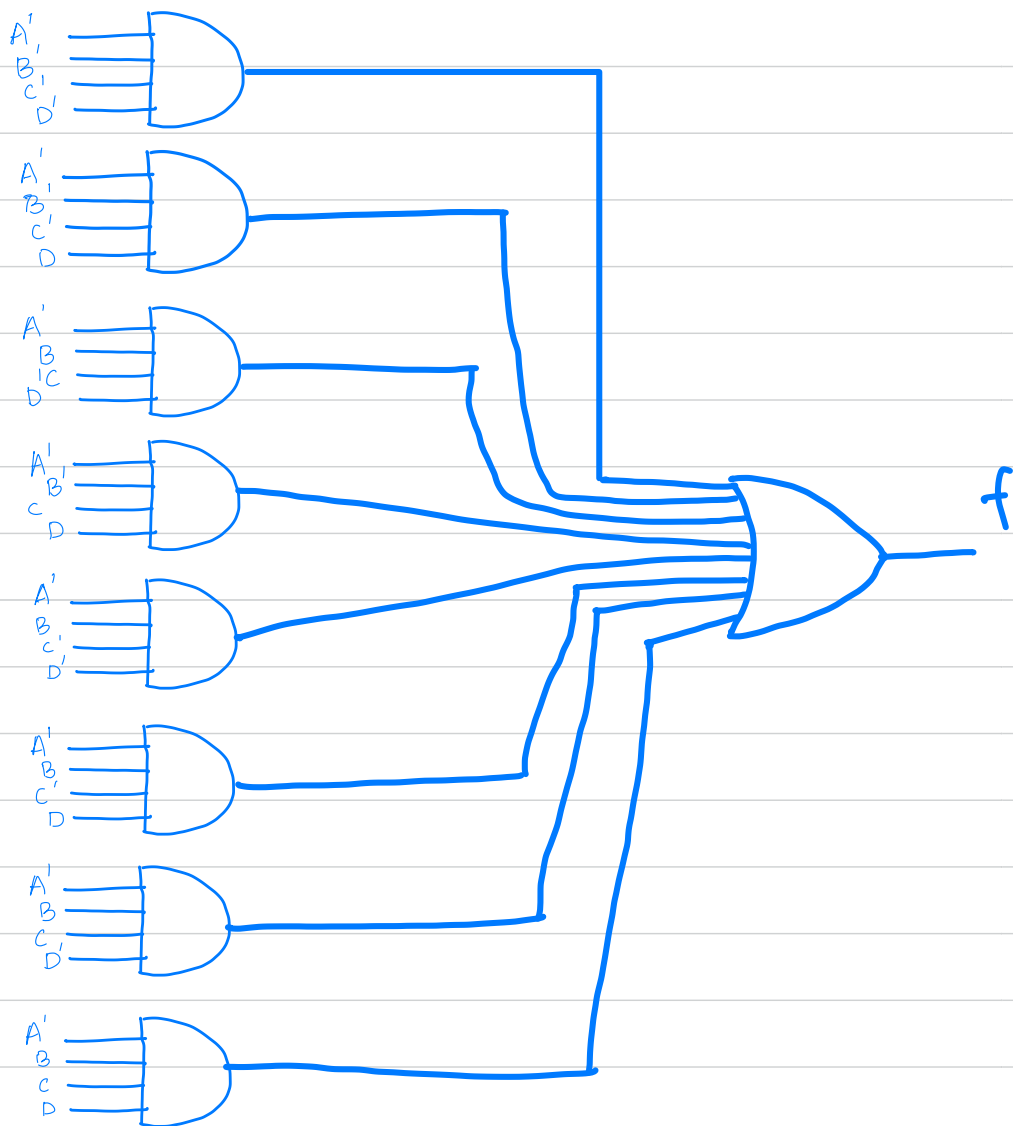
$$= \bar{A}\bar{D} + AC + \bar{C}\bar{D} + DB$$

$$= \sum m(0, 2, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15)$$

$$= \prod M(1, 3, 9)$$



q.	A	B	C	D	Output	
0	0	0	0	0	1	$A'B'C'D'$
1	0	0	0	1	1	$A'B'C'D$
2	0	0	1	0	1	$A'B'CD'$
3	0	0	1	1	1	$A'B'CD$
4	0	1	0	0	1	$A'BC'D'$
5	0	1	0	1	1	$A'BC'D$
6	0	1	1	0	1	$A'BCD'$
7	0	1	1	1	1	$A'BCD$
8	1	0	0	0	0	
9	1	0	0	1	0	
10	1	0	1	0	0	
11	1	0	1	1	0	
12	1	1	0	0	0	
13	1	1	0	1	0	
14	1	1	1	0	0	
15	1	1	1	1	0	



10. $E = 0$

	A	B	C	D	Output
0	0	0	0	0	0
1	0	0	0	1	0
2	0	0	1	0	0
3	0	0	1	1	0
4	0	1	0	0	0
5	0	1	0	1	1
6	0	1	1	0	1
7	0	1	1	1	1
8	1	0	0	0	1
9	1	0	0	1	1
10	1	0	1	0	1
11	1	0	1	1	1
12	1	1	0	0	1
13	1	1	0	1	1
14	1	1	1	0	1
15	1	1	1	1	1

$$E=1$$

	A	B	C	D	Output
0	0	0	0	0	1
1	0	0	0	1	1
2	0	0	1	0	1
3	0	0	1	1	1
4	0	1	0	0	1
5	0	1	0	1	1
6	0	1	1	0	0
7	0	1	1	1	0
8	1	0	0	0	0
9	1	0	0	1	0
10	1	0	1	0	0
11	1	0	1	1	0
12	1	1	0	0	0
13	1	1	0	1	0
14	1	1	1	0	0
15	1	1	1	1	0

$$E=0$$

AB		00	01	11	10
CD	00	0	0	1	d
	01	0	1	1	d
	11	0	1	1	d
	10	0	1	d	d

$$F_1 = AE' + BDE' + BCE'$$

AB		00	01	11	10
CD	00	1	1	d	
	01	1	1	d	
	11	1		d	
	10	1		d	d

$$E=1$$

$$F_2 = A'B'E + A'C'E$$

$$F = F_1 + F_2$$

$$= E'(A + B(C + D)) + EA'(B' + C')$$

